MILITARY SPECIFICATION SHEET

PENETRATOR, BULKHEAD, CONNECTORIZED,
FIBER OPTIC, 32 CHANNEL
(FOR INBOARD USE ON NAVY SHIPS AND SUBMARINES)

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of the Defense.

The complete requirements for acquiring the penetrator described herein shall consist of this specification sheet and the latest issue of DOD-P-24627(NAVY).

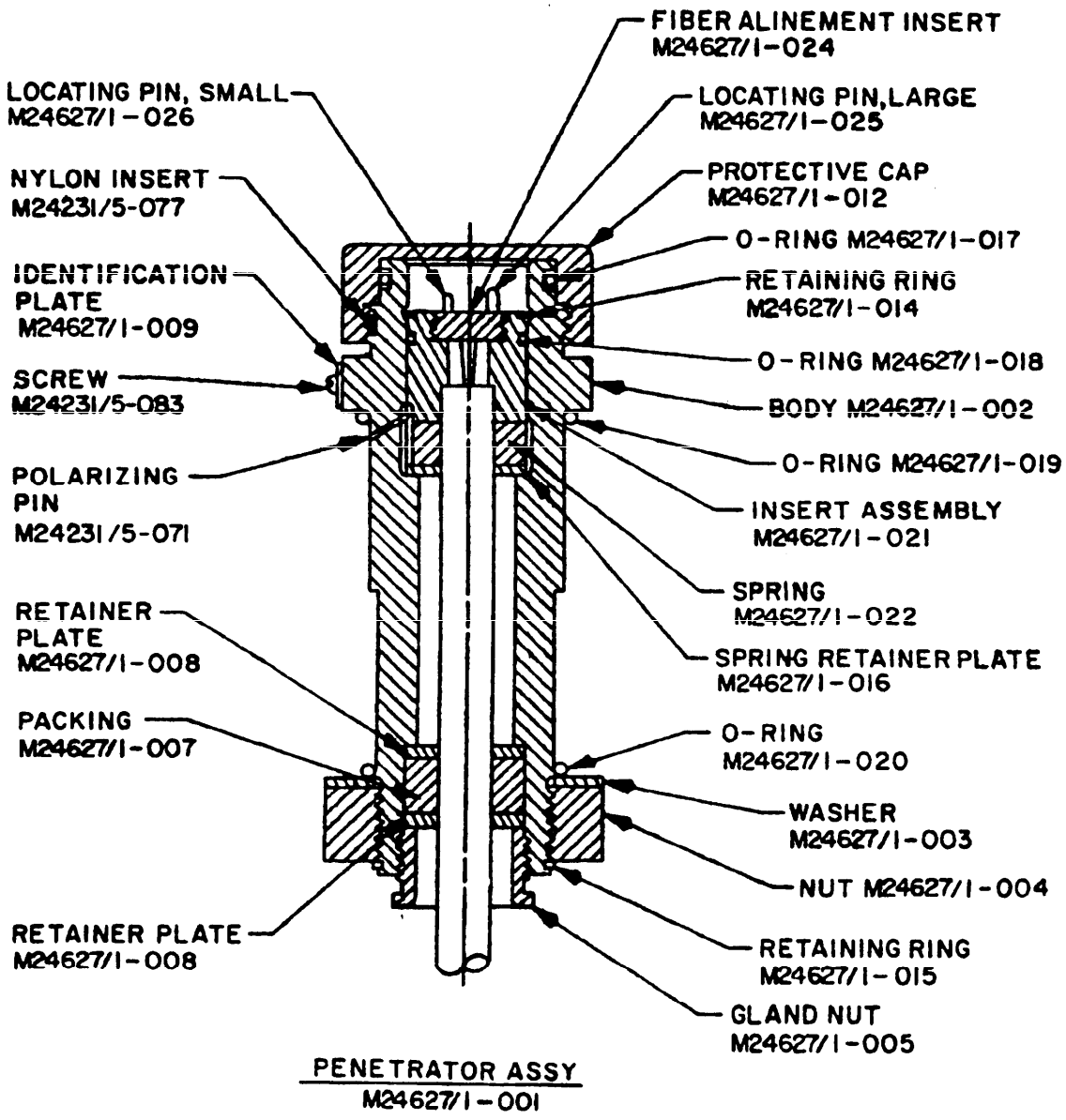


FIGURE 1. Dimensions and configurations.

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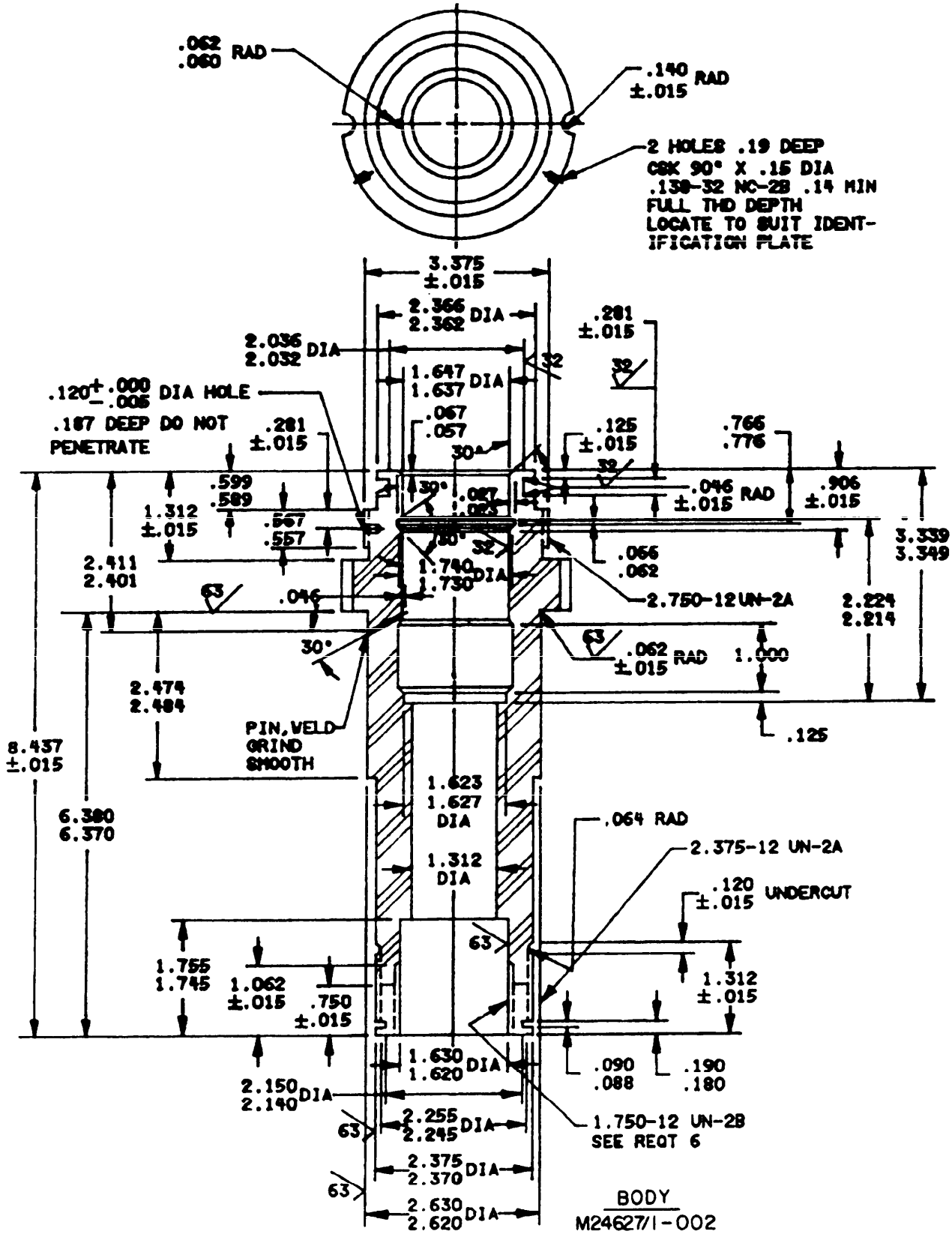
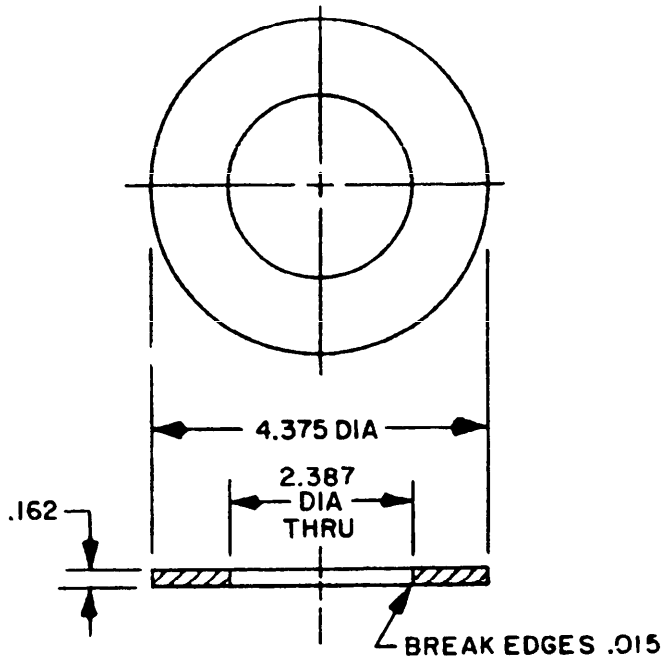
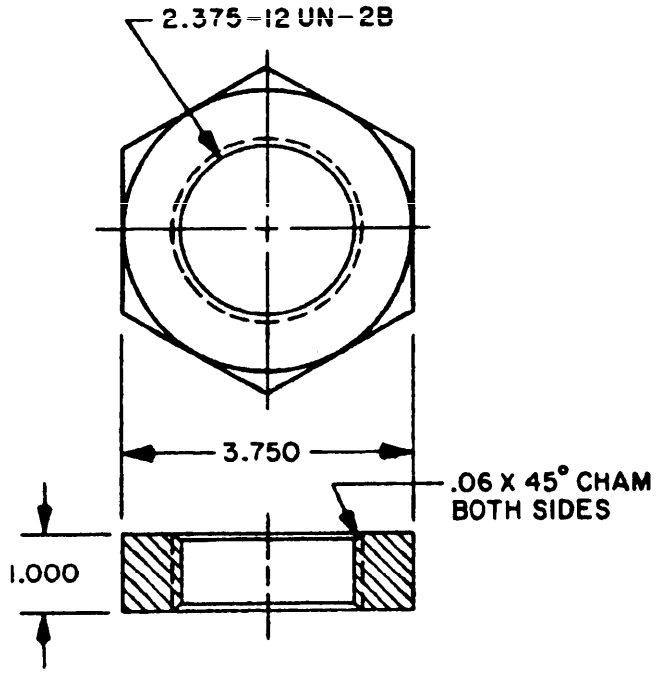


FIGURE 1. Dimensions and configurations - Continued.

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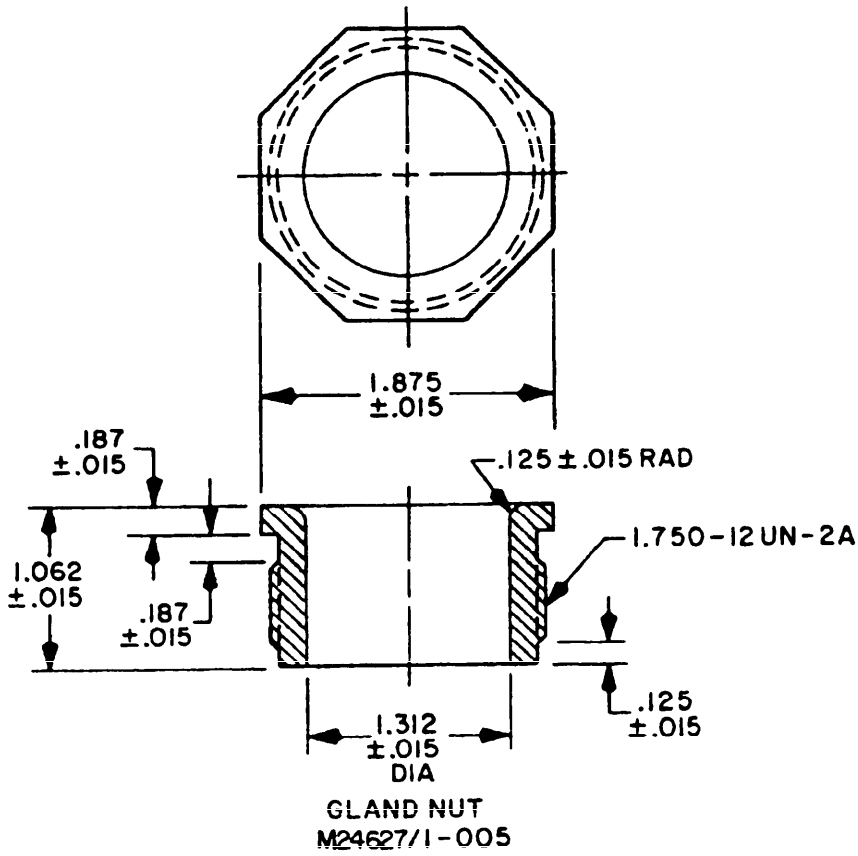
WASHER
M24627/1-003



NUT
M24627/1-004

FIGURE 1. Dimensions and configurations - Continued.

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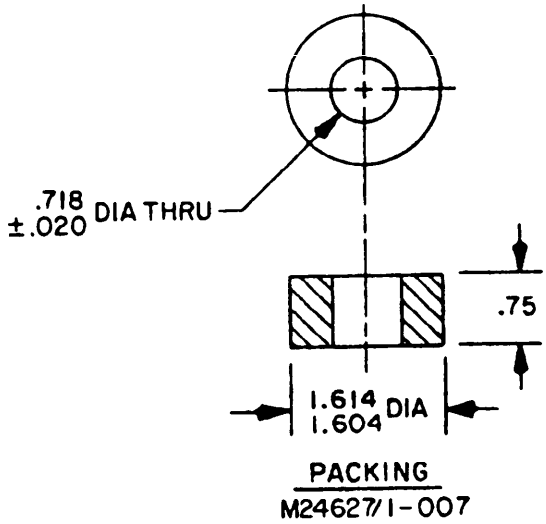
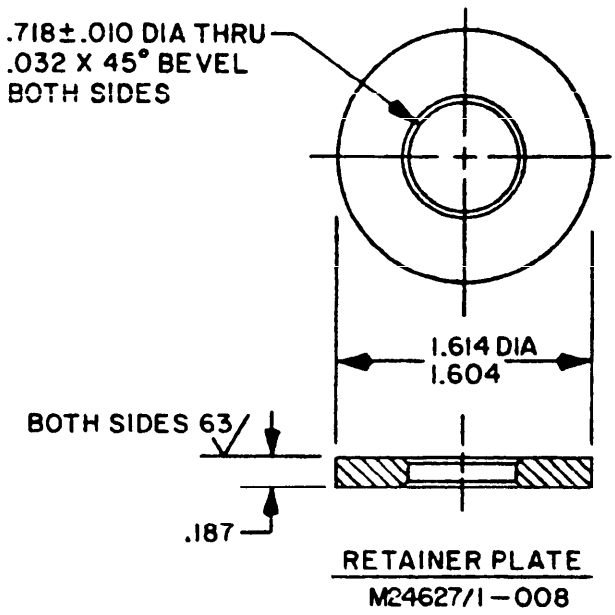


FIGURE 1. Dimensions and configurations - Continued.

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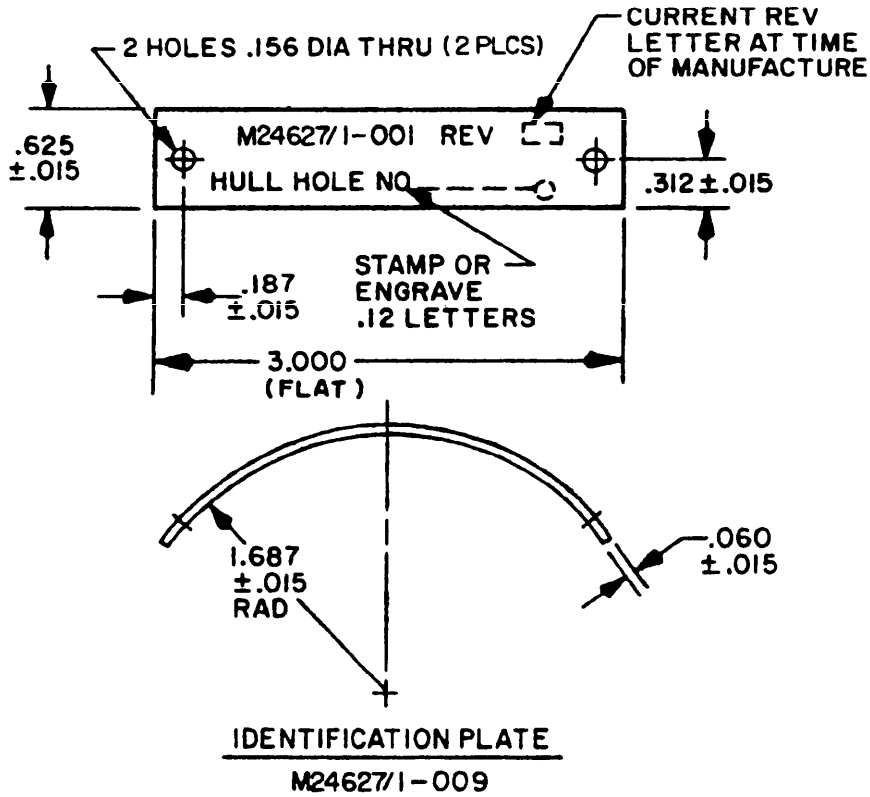


FIGURE 1. Dimensions and configurations - Continued.

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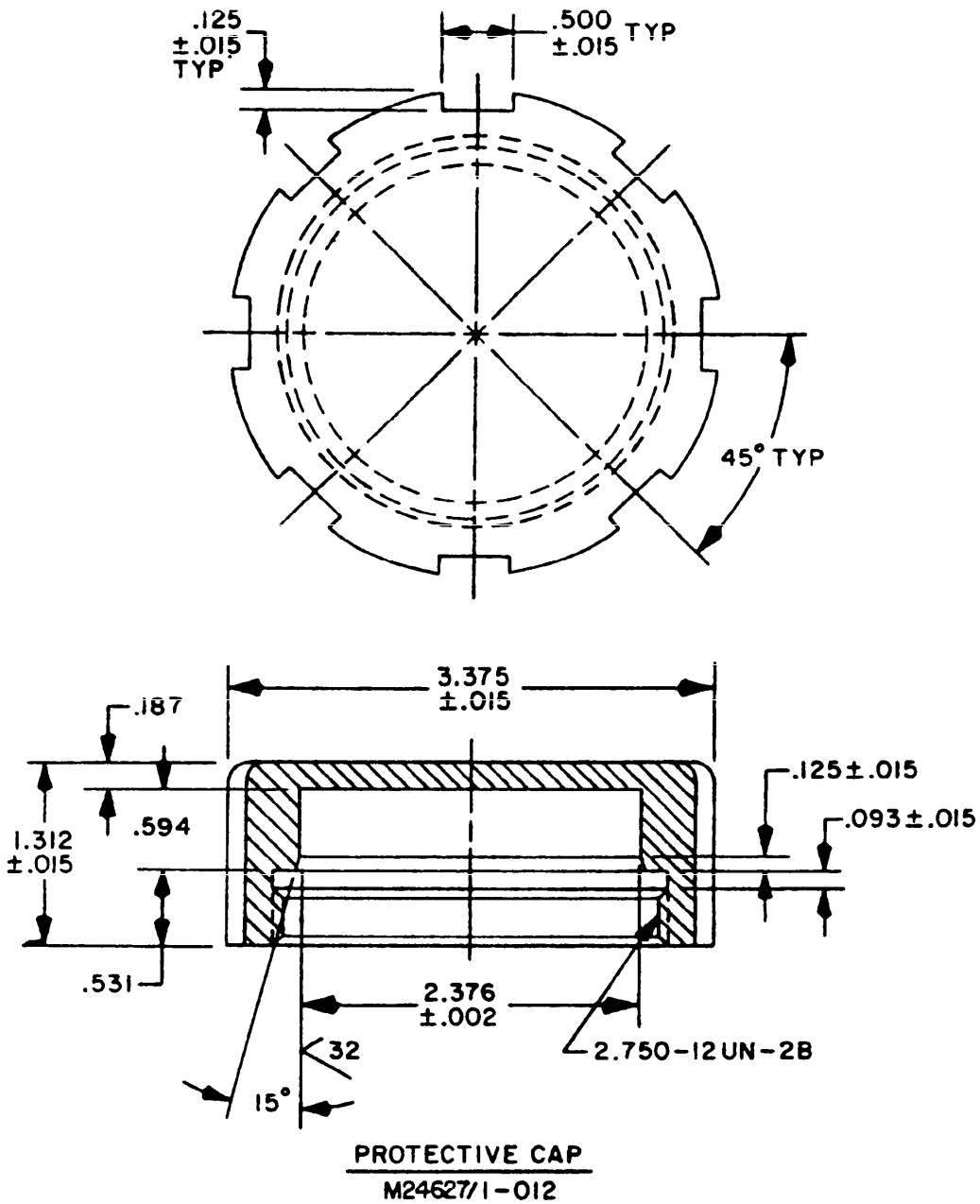
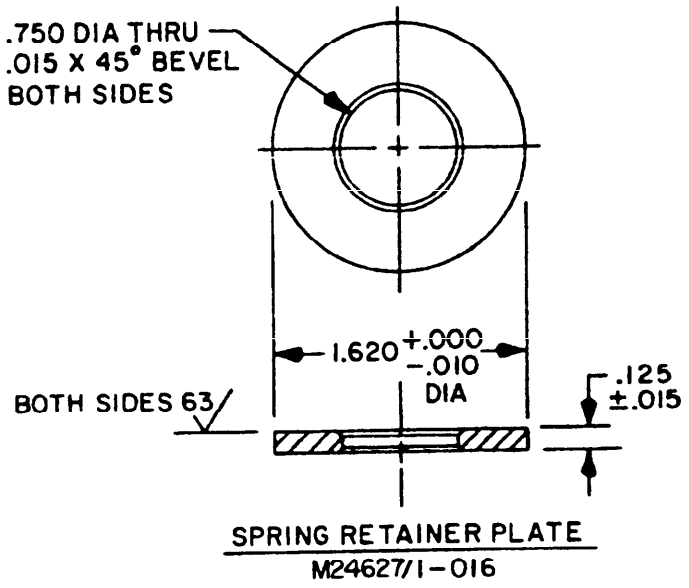


FIGURE 1. Dimensions and configurations - Continued.

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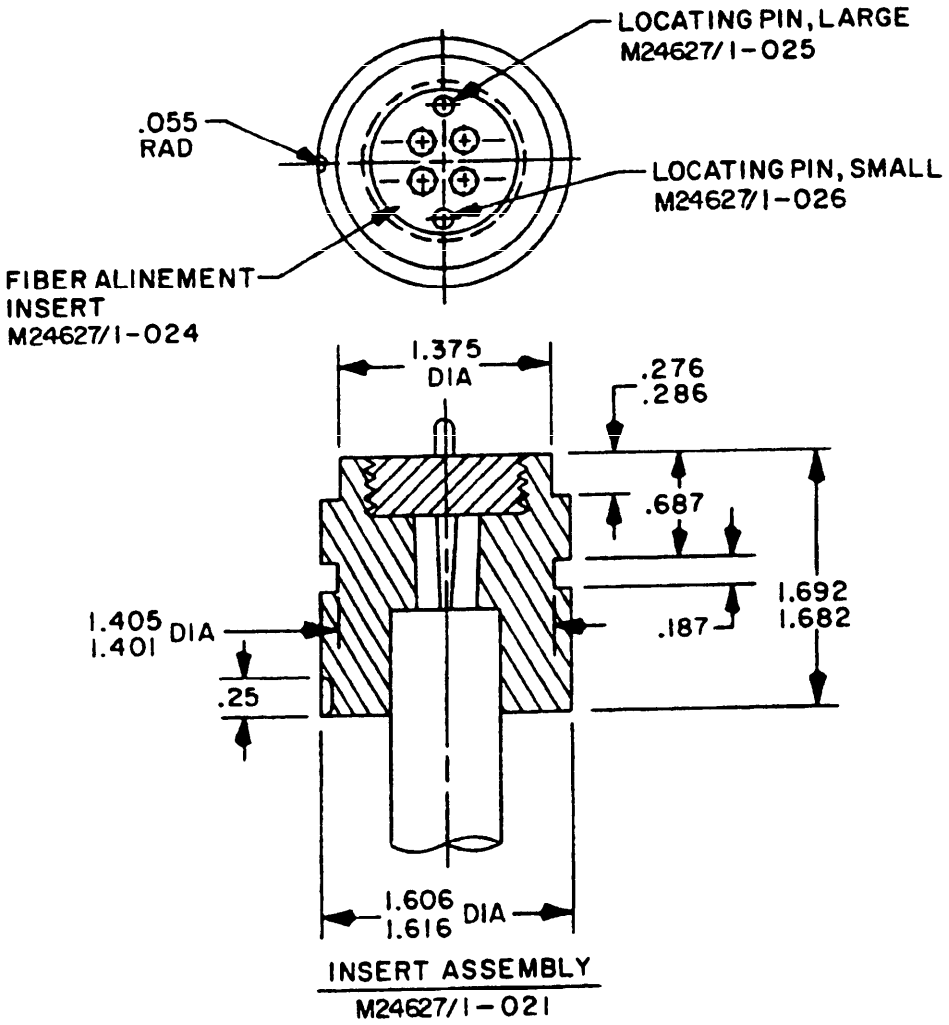


FIGURE 1. Dimenisons and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-19T23:30Z
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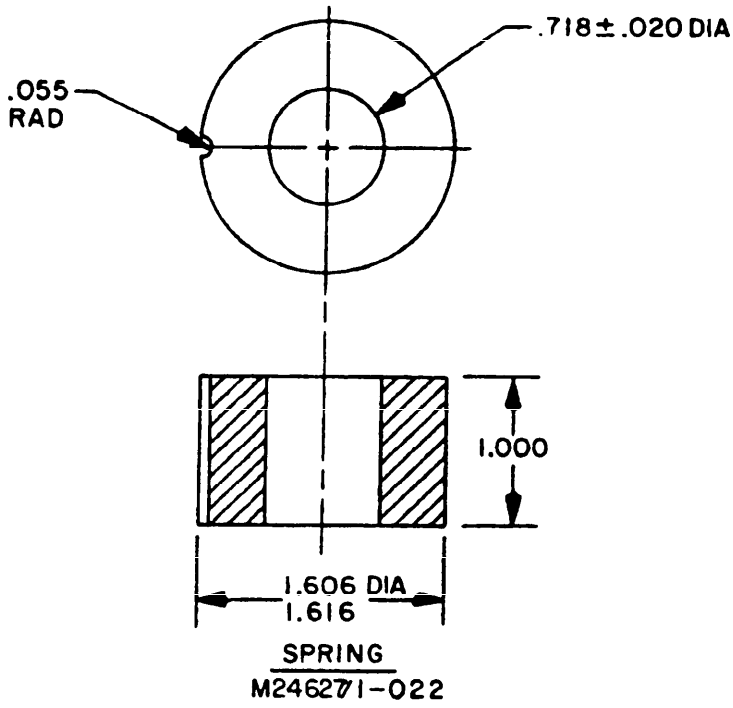
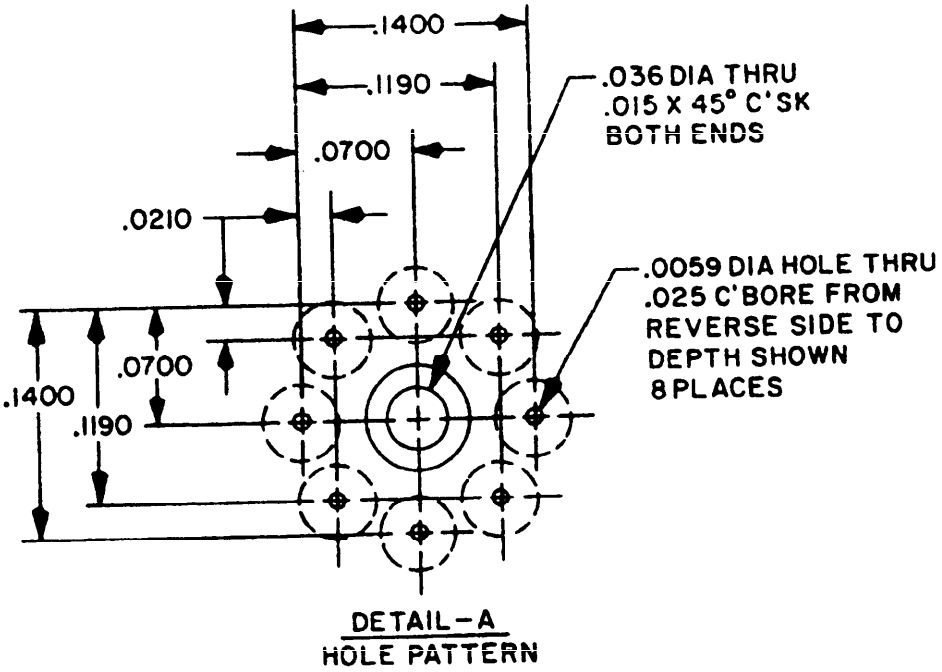
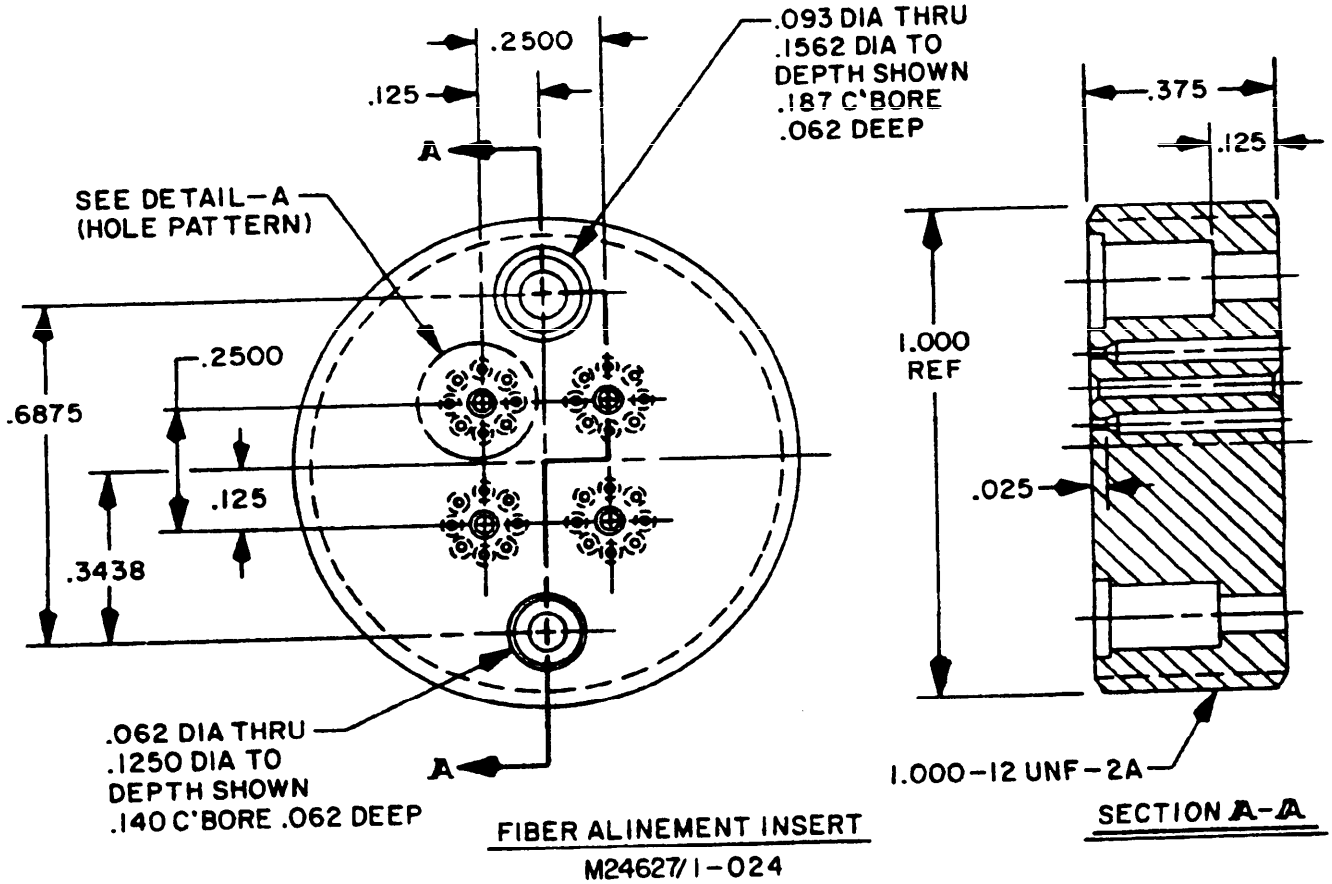


FIGURE 1. Dimensions and configurations - Continued.

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DETAIL-A
HOLE PATTERN

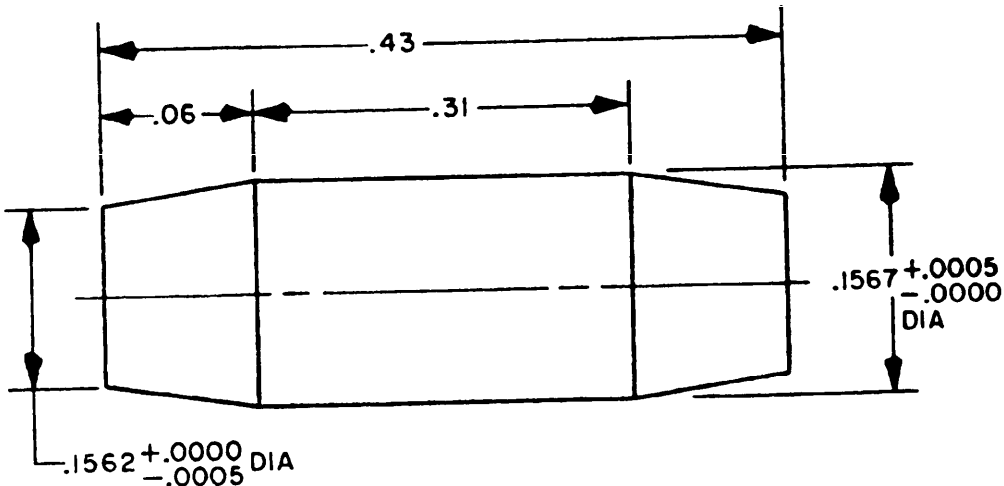


FIBER ALINEMENT INSERT
M24627/1-024

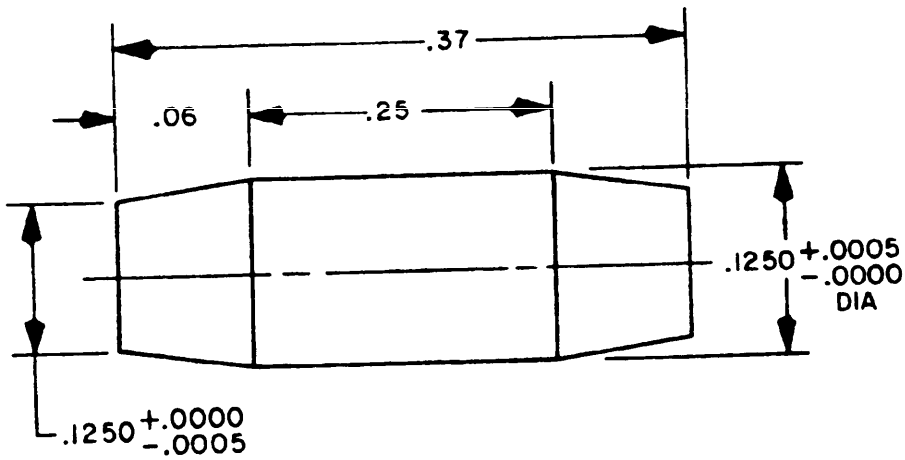
SECTION A-A

FIGURE 1. Dimensions and configurations - Continued.

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LOCATING PIN, LARGE
M24627/1-025



LOCATING PIN, SMALL
M24627/1-026

FIGURE 1. Dimensions and configurations - Continued.

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.0005	0.013	.1190	3.023	.500	12.7	1.606	40.79	2.245	57.02
.005	0.13	.120	3.05	.531	13.49	1.614	41.00	2.255	57.28
.0059	0.150	.125	3.18	.557	14.15	1.616	41.05	2.362	60.00
.015	0.38	.1250	3.180	.567	14.40	1.620	41.15	2.366	60.10
.020	0.51	.140	3.56	.589	14.96	1.623	41.22	2.375	60.32
.0210	0.533	.1400	3.560	.594	15.09	1.627	41.33	2.376	60.35
.023	0.58	.156	3.96	.599	15.21	1.630	41.40	2.387	60.63
.025	0.64	.1562	3.967	.625	15.88	1.682	42.72	2.401	60.99
.027	0.69	.1567	3.980	.687	17.45	1.687	42.85	2.411	61.24
.032	0.81	.162	4.11	.6875	17.463	1.692	42.98	2.474	62.84
.046	1.17	.180	4.57	.718	18.24	1.730	43.94	2.484	63.09
.055	1.40	.187	4.75	.750	19.05	1.740	44.20	2.620	66.55
.057	1.45	.190	4.83	.766	19.46	1.745	44.32	2.630	66.80
.06	1.5	.25	6.4	.776	19.71	1.750	44.45	2.750	69.85
.062	1.57	.2500	6.350	.906	23.01	1.755	44.58	3.000	76.20
.064	1.63	.281	7.14	1.000	25.40	1.875	47.63	3.339	84.81
.066	1.68	.31	7.9	1.062	26.97	2.032	51.61	3.349	85.06
.067	1.70	.312	7.92	1.312	33.32	2.036	51.71	3.375	85.73
.0700	1.78	.3438	8.733	1.375	34.93	2.140	54.36	3.750	95.25
.088	2.24	.37	9.4	1.401	35.59	2.150	54.61	4.375	111.13
.090	2.29	.375	9.53	1.405	35.69	2.214	56.24	6.370	161.80
.093	2.36	.43	10.9	1.604	40.74	2.224	56.49	6.380	162.05
								8.437	214.30

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.02 (0.51 mm) on two place decimals and ±.005 (0.13 mm) on three place decimals, angular tolerance is ±.30'.
4. 125√ Finish all over except as shown.

FIGURE 1. Dimensions and configurations - Continued.

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TABLE I. Parts list.

Part number	Description	Quantity	Remarks
M24627/1-001	Penetrator assembly	1	
M24627/1-002	Body	1	
M24627/1-003	Washer	1	
M24627/1-004	Nut	1	
M24627/1-005	Gland nut	1	
M24627/1-007	Packing	1	
M24627/1-008	Retainer plate	2	
M24627/1-009	Identification plate	1	
M24231/5-083	Screw	1	
M24231/5-077	Nylon insert	1	
M24627/1-012	Protective cap	1	
M24231/5-071	Polarizing pin	1	
M24627/1-014	Retaining ring	1	See requirement 3
M24627/1-015	Retaining ring	1	See requirement 4
M24627/1-016	Spring retainer plate	1	
M24627/1-017	O-ring, MS28775-329	1	
M24627/1-018	O-ring, MS28775-220	1	
M24627/1-019	O-ring, MS28775-335	1	
M24627/1-020	O-ring, MS28775-332	1	
M24627/1-021	Insert assembly	1	
M24627/1-022	Spring	1	
M24627/1-024	Fiber alinement insert	1	
M24627/1-025	Locating pin, large	1	
M24627/1-026	Locating pin, small	1	

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REQUIREMENTS

Design and construction

Retaining ring, part number M24627/1-014, shall be Spirolox (R) RR-165-S (Ramsey Corp.) or equal.

Retaining ring, part number M24627/1-015, shall be Truarc (R) 5100-225-W (Waldes Kohinoor) or equal.

The gland nut shall be tightened until the packing tightens against the fiber optic cables. This can be determined by moving the cables gently back and forth until the cables are gripped by the packing. The gland nut shall be turned approximately one fourth of a turn more to hold the cables in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this will damage the fiber optic cables.

Full or partial threads shall not extend more than 0.083 inch beyond the 3/4 inch thread length on the penetrator body.

See table I for parts information.

Welding of the pin, part number M24231/5-071, is allowed after the penetrator body has been age-hardened to achieve its required physical properties. This gives the manufacturing activity the option to make the body, nut, and washer from any age hardened NI-CU-AL alloy in accordance with class A of QQ-N-286 that will meet the mechanical properties required in this specification. In that case, the chemical and physical certification obtained from the manufacturer of the NI-CU-AL alloy will satisfy certification requirements when supplimented by the required traceability markings identifying the finished product to the bar stock on which the chemical and physical analysis was performed.

The M24627/1-001 Fiber Optic Penetrator assembly is designed for installation in the MIL-C-24231/25-002 submarine hull insert assembly.

Custodians
Navy - SH

Review activities:
Navy - AS, EC
DLA - ES

User activities:
Navy - CG, MC

Agent:
DLA - ES

Preparing activity:
Navy - SH

(Project 6060-N012-01)

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